WEEK 1 DELIVERABLE REPORT: SELECTED TOPIC + INITIAL PROBLEM STATEMENT + AUDIT LOG

Precision Agricultural Drone Spraying Coordination Platform for the Mekong Delta (AgriUAV Platform)

Project Metadata & Overview

Course	ITA301 — Information System Analysis & Design
Academic Term	Summer 2026
Deliverable Item	W1 — Topic Selection + Initial Problem Statement + AI Audit Log v0 (Due: 2026-05-30)
System Domain	Agricultural UAV — Precision Drone Spraying Coordination in the Mekong Delta
Course Instructor	Mr. Tran Thanh Nguyen
Project Team & Roles	• Dang Thanh Cong — Project Manager • Nguyen Thi Thanh Thao — Business Analyst • Le Van Minh — System Analyst • Tran Anh Khoa — System Designer • Dang Vo Thanh Hieu — QA / Presenter
Document Version	v0 — Official W1 Final Document

1. Selected Topic Specification

Following the course requirements and preliminary ecosystem evaluation, the project team selected a specific domain under the Information Systems core guidelines. The details of the defined system topic are structured below:

System Name	AgriUAV Platform
Full Project Title	Precision Agricultural Drone Spraying Coordination Platform for the Mekong Delta
Sub-domain Classification	Agricultural UAV (Precision dispatching, fleet monitoring, and scheduling optimization)
Topic Number	Topic #2 (Selected from the approved course sample topic guidelines)

2. Business Context & Domain Overview

The Vietnamese Mekong Delta (ĐBSCL) is the country's primary rice-producing region, covering approximately 1.5 million hectares of cultivated rice per season. Agricultural drone services have emerged as a critical operational layer in this region, yet the digital infrastructure required to manage these services at scale remains entirely absent.

2.1. Ecosystem Analysis — 3-Layer Model

The Agricultural UAV ecosystem in Vietnam is structured across three distinct layers. This decomposition reveals the precise market gap that AgriUAV Platform is designed to fill:

Ecosystem Layer	Current Market State	AgriUAV Platform Intervention
Hardware Layer	Mature and saturated — DJI Agras (~80% Vietnam market share), XAG (~15%), and domestic brands (~5%). Hardware availability is not a bottleneck.	Integration via mobile GPS at MVP stage. Planned MAVLink telemetry protocol support in future versions.
Service Layer	Local cooperatives (HTX) and private enterprises provide commercial spraying services at ~150,000 VND/ha. Operations are entirely manual — managed via Zalo, Excel, and paper logbooks.	Provides the end-to-end digital coordination layer to automate scheduling, dispatch, and compliance for existing service providers.
Platform Layer	COMPLETELY VACANT — No dedicated, localized IS for operational management, pilot dispatch, scheduling, or regulatory compliance logging currently exists in Vietnam.	AgriUAV Platform fills this strategic gap, serving as the enterprise-grade operational coordination system for drone service cooperatives.

Conclusion: The Hardware Layer and Service Layer are already occupied by established players. The Platform Layer remains entirely open — this is the market gap that an Information System is positioned to address, and is precisely aligned with the scope of ITA301.

2.2. Key Stakeholder Overview

The system serves four primary stakeholder groups, each with distinct operational needs:

- Cooperative Management (HTX): Fleet scheduling, dispatch optimization, regulatory reporting, and operational cost control.
- Drone Operators / Pilots: Digital field boundary access, GPS-guided navigation, automated flight logging, and real-time safety alerts.
- Rice Farmers: Simplified spray service booking, real-time drone status tracking, and pesticide usage verification.
- Regulatory Bodies (Plant Protection Dept. / Operations Dept.): Compliance reporting, drone ID tracking, and pilot certification verification per Decree 288/2025/ND-CP.

2.3. Why This System Is Needed Now

Two concurrent pressures create an urgent deployment window for this platform:

- Decree No. 288/2025/ND-CP (effective November 05, 2025): Legally mandates digital recording of unique drone identification numbers and valid pilot certifications for all commercial UAV operations — making manual paper-based logging legally non-compliant.
- Operational Cost Crisis: Empirical research (Nguyen et al., 2026; N=940 surveyed farms) confirms that current manual scheduling methods inflate operational costs by 15–20% per season through duplicate bookings, deadhead flights, and unverified pesticide dispensing.

3. Initial Problem Statement

Agricultural cooperatives (HTX) and drone service enterprises in the Mekong Delta currently face a critical absence of centralized digital tools to manage, schedule, and monitor their spraying drone fleets. This operational gap produces five categories of measurable deficiencies:

3.1. High Operational Costs

Managing drone spraying requests manually via Zalo text groups, scattered Excel sheets, or paper notebooks causes persistent schedule conflicts, overlapping shifts, and heavily inflated deadhead flights (empty drone transit flights between fields), driving operational overhead up by 15–20% per crop season (Nguyen et al., 2026).

3.2. Low Field-Level Spraying Efficiency

Without real-time digital field boundary mapping or precise GPS parcel tracking, drone operators spray entire land areas uniformly rather than applying targeted, variable pest dosages to heavily infected crop zones. This results in excessive agrochemical wastage and localized yield drops in under-treated areas.

3.3. Absence of Operational Audit Trails

Cooperatives have no automated telemetry validation mechanisms. Drone operators may under-dose chemical mixtures or inflate recorded flight hours to claim fuel and compound allowances without detection, as no verifiable digital audit trail exists for management review.

3.4. Severe Flight Safety Hazards

The complete absence of automated geofencing exposes surrounding infrastructure to extreme safety incidents. In Long An Province, an unmonitored agricultural drone collided with a 110kV high-voltage power transmission line, triggering a grid failure that left over 76,000 rural customers without electricity (Dan Viet, 2024). Separately, off-target chemical drift from improperly monitored drones has caused agrochemical exposure incidents affecting neighboring farmland (Vietnam Naval News, 2024).

3.5. Regulatory Non-Compliance

Manual flight logging fails to satisfy the requirements of Decree No. 288/2025/ND-CP, which mandates the systematic digital recording of unique drone identification numbers and valid operator certifications for all commercial light-UAV operations in Vietnam. Non-compliance exposes cooperatives to legal liability and operational suspension risk.

4. System Scope

4.1. In-Scope — System Functions

- Drone fleet management: drone identification codes, model specifications, tank capacity, and operational status (Available / In-Flight / Under Maintenance)
- Pilot profile management: certifications, licenses, and flight history records compliant with Decree 288/2025/ND-CP
- Spray scheduling: farmers and HTX managers submit spray requests; system confirms and records assignments
- Automated drone dispatch: assignment logic based on battery level, current GPS position, and schedule availability
- Real-time spray progress tracking on digital field map: drone position overlay and percentage area completion
- Automated alerts: low pesticide level, low battery, adverse weather (wind speed > 3 m/s), and geofencing boundary breach
- Pesticide inventory management and spray formula records per approved chemical registry
- Spray log export and seasonal compliance reporting for the Plant Protection Department
- Role-based access control: Admin / HTX Manager / Operator / Farmer
- System scale: maximum 50 drones per cooperative, within rice cultivation zones in the Mekong Delta

4.2. Out-of-Scope — Explicitly Excluded

- Direct drone hardware control from software (fly-by-wire or autopilot command interfaces)
- AI-based pest detection from drone imagery (computer vision or deep learning models)
- Integration with irrigation systems or other agricultural machinery
- Financial management, accounting, or VAT invoicing modules
- Online flight permit application with the Operations Department (UTM system integration)
- Physical cloud deployment or real-hardware drone integration (outside ITA301 scope — design only)

5. AI Artifact Audit Log (Week 1 — v0)

This audit log documents the iterative interaction, prompt engineering, and strategic human validation (Human Delta) between the project team and AI models during Week 1. Each entry records the full rationale for accepting, rejecting, or refining AI-generated outputs.

No.	DTC Component	AI Tool	Prompt (Full Context)	AI Output (Summary)	Decision & Human Delta	Verification
01	**Problem & Business Decomposition**	Claude	"We are developing an IS project for ITA301 on Agricultural UAVs in the Mekong Delta. Suggest 3 distinct sub-domain directions, stating the primary business problem and implementation complexity for each."	3 directions: (1) Fleet Management — generic tracking; (2) Precision Spraying Platform — localized scheduling; (3) Marketplace — hourly drone rentals. All rated ★★★ complexity.	☑ SELECTED (2) Precision Spraying Platform: • Empirical data: Nguyen et al. 2026, N=940. • Pain points validated: HTX uses Zalo/Excel. • Platform layer is vacant in ecosystem. ✕ REJECTED (3): Scope too broad, no IS depth. ✕ REJECTED (1): Overlaps with another team's topic.	Ecosystem gap confirmed via Lao Dong (2025): 'digital scheduling problem remains unresolved in ĐBSCL.'
02	**Pattern Recognition & Comparison**	ChatGPT	"Compare Agricultural UAV, Delivery UAV, and Surveillance UAV across: (a) stakeholder complexity, (b) regulatory burden in Vietnam, and (c) data availability for an IS academic project."	AI concluded Delivery UAV is 'easiest' due to lower perceived regulation. Agricultural UAV ranked medium complexity.	⚠ AI RANKING REJECTED — incorrect evaluation: Agricultural UAV has a more structured legal framework (Decree 288/2025/ND-CP + Plant Protection Dept) than Delivery UAV.	Decree 288/2025/ND-CP cross-checked via thuvienphapluat.vn — confirms specific drone ID and pilot certification logging requirements for agricultural operators.

No.	DTC Component	AI Tool	Prompt (Full Context)	AI Output (Summary)	Decision & Human Delta	Verification
					☑ Maintained Agricultural UAV selection: (a) peer-reviewed research available (Nguyen 2026), (b) verifiable field incidents, (c) richer stakeholder map for IS analysis.	
03	**Decomposition — Ecosystem Mapping**	Claude	"For the AgriUAV Platform (ITA301, Mekong Delta): decompose the current Agricultural UAV ecosystem into structural layers. Identify which layers are occupied and which represent market gaps for an IS solution."	AI proposed a 2-layer split: Hardware/Device layer and Application/User layer. Suggested the gap lies in 'application integration.' No distinct Service Layer or Platform Layer identified.	✕ REJECTED 2-layer model — oversimplified and inaccurate. ☑ Team defined 3-Layer Model: • Hardware Layer (DJI/XAG — occupied) • Service Layer (HTX fleets — occupied, manual) • Platform Layer (digital IS coordination — VACANT) This reflects actual business reality confirmed by field reporting.	3-layer model validated: Thanh Nien (June 2025) confirms 1,215 active drones across 3 provinces with no platform-layer management software reported.

6. Verified Academic & Regulatory References

[1] Nguyen, D.L. et al. (2026). "The Effect of Farm Size on the Decision to Adopt Digital Technology: The Case of Unmanned Aerial Vehicles in Rice Production in Vietnam." Research on World Agricultural Economy. https://journals.nasspublishing.com/index.php/rwae/article/view/2358

[2] Government of Vietnam. Decree No. 288/2025/ND-CP — Regulatory Framework for Unmanned Aerial Vehicle Flights and Light-UAV Digital Registry (Effective: November 05, 2025). https://thuvienphapluat.vn/van-ban/Giao-thong-Van-tai/Nghi-dinh-288-2025-ND-CP-quan-ly-tau-bay-khong-nguoi-lai-va-phuong-tien-bay-khac-679996.aspx

[3] Lao Dong Newspaper. (2025). Digital Transformation Gaps in Mekong Delta Cooperatives: The Unresolved Software Problem. https://laodong.vn/xa-hoi/drone-nong-nghiep-va-bai-toan-so-hoa-o-dbscl-1498970.ldo

[4] Dan Viet Digital News. (2024). Investigation Report: Agricultural Drone Incident Involving 110kV Transmission Line Contact, Causing Outages for 76,000 Customers in Long An Province. https://tuoitre.vn/drone-phun-thuoc-tru-sau-va-vao-day-110kv-gay-mat-dien-o-5-huyen-tai-long-an-20241014173928997.html

[5] Vietnam Naval News. (2024). Management of Unmanned Aerial Systems and Flight Permit Processes via the Operations Department. https://baohaiquanvietnam.vn/tin-tuc/quan-ly-thiet-bi-bay-khong-nguoi-lai-trong-nong-nghiep-khong-de-cong-nghe-tro-thanh-hiem-hoa

[6] Thanh Nien Newspaper. (June 2025). Province-Level Statistical Breakdown of Active Agricultural Drone Fleets in the Mekong Delta. https://thanhnien.vn/bung-no-uav-nong-nghiep-o-mien-tay-185250617232615926.html

[7] Vietnam News. (2026). Use of Agricultural Drones on the Rise in the Mekong Delta. https://vietnamnews.vn/society/1688198/use-of-agricultural-drones-on-the-rise-in-the-mekong-delta.html